

From Future Danger to Communication Value: Risk-Aware Visual Communication for Robot Navigation

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Abstract—A bandwidth-limited robot cannot communicate every camera update, yet an early warning of danger does not identify when a packet is worth spending. We study this distinction through a causal Webots pipeline that combines deterministic ego-motion rollout, decision-level visual Safety Value, and an interpretable feasibility-deterioration precursor. In an independently sealed evaluation of 240 physical configurations, future-motion-conditioned danger raises 2-s AUPRC from 0.9063 to 0.9937 (difference 0.0874, 95% CI [0.0772, 0.0985]). Without refitting, the precursor anticipates 16 of 19 Safety-Value onsets across four new physical families with 0.576-s median lead while active for 1.465% of time. We then compare SEND NOW with preserving the same packet using identical-prefix, exactly 72,000-byte branches. Thirty direct opportunities are helpful in 13 cases, harmful in 7, and neutral in 10. After repairing the target to adjacent opportunities, 45 comparisons contain 11/15/19 helpful/harmful/neutral cases, but the best leave-family-out causal model reaches only 0.628 helpful AUPRC and fails its frozen selection gate. Thus future danger, decision relevance, and immediate transmission utility are empirically distinct; no safer scheduler is claimed.

I. INTRODUCTION

Visual observations support obstacle avoidance, but transmitting them at the camera rate is often incompatible with constrained or intermittent links. The central scheduling question is therefore not merely whether an image contains important content; it is which update deserves one of the remaining communication opportunities. Task-oriented image coding and collaborative perception address rate–task trade-offs [1], [2], while learned communication systems decide where, when, or with whom agents exchange features [3], [4]. For closed-loop navigation, however, “dangerous,” “decision-relevant,” and “useful to send now” need not describe the same instant.

Consider a robot whose present motion makes a later obstacle encounter predictable. Future risk may rise before the obstacle produces a materially new camera observation. Later, a fresh frame may change the receiver’s feasible motion set, but sending it immediately can still be suboptimal if only a few packets remain and a more consequential transition follows. This creates a temporal hierarchy: future danger is an early physical forecast; decision value asks whether fresh visual information changes safe planning; transmission utility asks whether consuming a packet now improves the eventual closed loop.

We investigate this hierarchy through a sequence of frozen, causal experiments (Fig. 1). The sender and receiver use only current or past decoded visual information and robot-local

state. Physical contact, true clearance, and future simulator states remain evaluator-only. Real serialized packet bytes are counted, and competing closed-loop policies receive the same episode budget.

This paper makes exactly three contributions:

- 1) A causal future-danger formulation based on deterministic ego-motion rollout, independently confirmed on a sealed set of 240 physical configurations.
- 2) A decision-space definition of visual Safety Value and a frozen, cross-scenario feasibility precursor that anticipates 16/19 onsets with 0.576-s median lead while active only 1.465% of the time.
- 3) An exact matched-wire counterfactual study that directly measures helpful, harmful, and neutral packet opportunities, exposes narrow utility windows, and reports a bounded negative result when causal utility models fail cross-family selection gates.

II. RELATED WORK

A. Task-Oriented Robotic Communication

Task-oriented image communication jointly considers rate, reconstruction, and downstream distortion rather than pixel fidelity alone [1]. TAGIC similarly uses task context to guide image communication for simulated teleoperation [5]. Multi-agent perception methods learn when to establish communication groups [3], transmit spatially sparse high-confidence features [2], or exchange compressed features for joint perception and forecasting [4]. These systems demonstrate that task relevance can reduce communication, but their primary endpoints are reconstruction, perception, or forecasting. Our endpoint is matched-cost closed-loop navigation, and our variable of interest is the temporal marginal value of one visual packet.

B. Predictive Risk and Information Timing

Predictive attention can place visual processing on regions likely to affect future control and unsafe-condition detection [6]. Risk-aware spatio-temporal corridors likewise convert predicted environmental evolution into constraints for motion planning [7]. Our work does not claim to introduce task-oriented communication, predictive attention, or risk-aware planning. It instead tests whether future physical danger, receiver-side decision relevance, and immediate packet utility coincide. The matched-wire intervention reveals that they do not.

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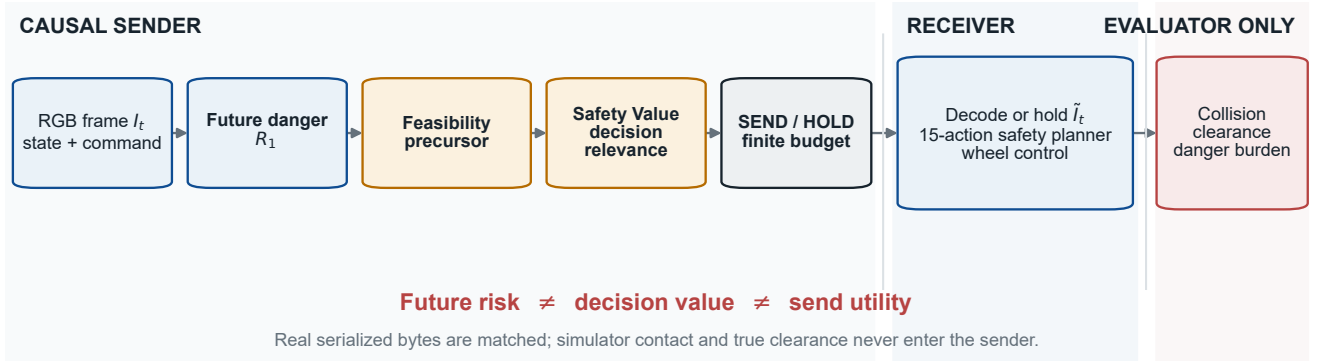


Fig. 1. Causal study design. Predicted danger can precede a deterioration of the receiver’s feasible action set; Safety Value confirms that a fresh frame changes the safety decision. Neither signal alone establishes that spending a finite packet immediately is better than holding it for a later observation.

III. PROBLEM FORMULATION

At control step t , the sender observes robot-local state s_t , the issued wheel command u_t , and camera frame I_t . The receiver controls from the most recently decoded frame $\tilde{I}_t = I_{\ell(t)}$, where $\ell(t) \leq t$ is the last send time. A causal policy has information

$$\mathcal{I}_t = \{s_\tau, u_\tau, I_\tau, a_\tau : \tau \leq t\} \cup \{\tilde{I}_t, b_t\}, \quad (1)$$

where b_t is remaining wire budget and $a_t \in \{\text{send}, \text{hold}\}$. It cannot access future frames, future physical states, contact, or evaluator geometry.

Let $\Pi(I, s)$ be a frozen safety-first local planner. We distinguish three quantities. Future danger $R_{1,t}$ scores whether the current motion is headed toward low clearance. Decision value V_t indicates whether planning with I_t rather than $\tilde{I}_t$ produces a material safety transition. Immediate transmission utility is the counterfactual closed-loop benefit of choosing send instead of hold while preserving future budget consequences. The last quantity is not assumed equal to either $R_{1,t}$ or V_t .

If a serialized packet costs $c = 24,000$ bytes including codec and protocol overhead, all primary Q6 policies obey

$$\sum_t c \mathbb{I}[a_t = \text{send}] = 72,000 \text{ bytes/episode}. \quad (2)$$

The receiver decodes each transmitted packet and holds that reconstruction between sends. Thus equal nominal rates are insufficient: Eq. (2) is checked from actual wire records.

IV. METHOD

A. Predictive Ego Danger

The motion predictor is deterministic differential-drive kinematics, not a learned model. For wheel rates (ω_L, ω_R) , wheel radius r_w , and axle length L , the twist is

$$v = \frac{r_w}{2}(\omega_R + \omega_L), \quad \Omega = \frac{r_w}{L}(\omega_R - \omega_L), \quad (3)$$

and is integrated exactly as a constant twist over 32-ms increments. The state-only rollout holds measured (v, Ω) constant. The command-conditioned rollout instead converts the issued wheel-command schedule to a piecewise-constant twist. Their all-stable ADEs at 0.5/2.0 s are $1.20561 \times 10^{-4} / 7.15992 \times 10^{-4}$

m (state-only) and $6.398 \times 10^{-6} / 1.3655 \times 10^{-5}$ m (command-conditioned).

For each polyline $\Gamma_{k,t}$ and currently represented obstacle set $\mathcal{O}_t$, define physical-footprint clearance

$$d_{k,t} = \min_{p \in \Gamma_{k,t}, o \in \mathcal{O}_t} \text{dist}(p, o) - r_b, \quad R_{k,t} = -d_{k,t}, \quad (4)$$

with body radius $r_b = 0.037$ m. R_0 uses the current center point; R_1 uses the state-only 2-s rollout and is the primary downstream risk signal; R_2 uses the command schedule and is secondary/mechanistic. This naming follows the frozen M9 implementation.

B. Decision-Level Safety Value and Precursor

The receiver evaluates 15 local actions: three forward speeds crossed with five angular rates. Each 1.5-s trajectory has conservative minimum clearance $m_{i,t}$ from decoded visual obstacle estimates, including range uncertainty. It is hard-feasible at $m_{i,t} \geq h = 0.025$ m and preferred-safe at $m_{i,t} \geq p = 0.075$ m. Selection prioritizes preferred-safe moving actions, then hard-safe maximum-margin actions, and finally stop.

Safety Value compares the held-image plan with a shadow plan from the current causal image. It triggers if any frozen condition holds: (i) selected safety class deteriorates; (ii) a nonempty moving-safe set collapses; (iii) the held action changes and is no longer hard-feasible; or (iv) at least one third of held-safe actions disappear and the held action’s matched margin falls by at least 0.01 m (transition from infinite to finite margin also counts as a large drop). A Q5 onset must additionally occur after step 7, before contact, and retain at least two current safe alternatives. This is decision-level value, not an arbitrary path change or pixel difference.

The precursor uses the exact soft feasibility mass of the current plan. With $N = 15$, $s = p - h = 0.05$ m, and finite replacement $\bar{m}_{i,t} = m_{i,t}$ if finite or $p + 20s$ otherwise,

$$z_{i,t} = \text{clip}\left(\frac{\bar{m}_{i,t} - h}{s}, -60, 60\right), \quad (5)$$

$$\Phi_t = \frac{1}{N} \sum_{i=1}^N \frac{1}{1 + e^{-z_{i,t}}} \in [0, 1]. \quad (6)$$

If every margin is non-finite, the implementation sets $\Phi_t = 1$. For the eight past-through-current samples $y_j = \Phi_{t-\tau+j}$ at $\tau_j = 0.032j$ s, the causal OLS slope is

$$\beta_t = \frac{\sum_{j=0}^7 (\tau_j - \bar{\tau})(y_j - \bar{y})}{\sum_{j=0}^7 (\tau_j - \bar{\tau})^2}. \quad (7)$$

No future sample enters β_t . With the exact frozen constants

$$\begin{aligned} \phi^* &= 0.6543448254639964, \\ \beta^* &= -0.11876628431105299 \text{ s}^{-1}, \end{aligned}$$

the precursor is

$$P_t = \mathbb{I}[\Phi_t \leq \phi^* \wedge \beta_t < \beta^*]. \quad (8)$$

C. Causal Matched-Budget Probe

The closed-loop probe sends three 24-kB decoded-image packets. U_0 uses the fixed schedule $\{0, 109, 218\}$. A_0 uses current clearance risk R_0 in the same causal scheduling stack. The final precursor policy uses R_1 , requires two consecutive $P_t = 1$ samples, and spends its middle packet when that persistent opportunity is available; otherwise it falls back at step 217. A startup packet at step 0 and protected reserve at step 218 are fixed. A precursor opportunity remains valid for 25 steps. Codec, decoded perception, planner, controller, physical cell, and total bytes are shared. This policy is an intervention probe, not a selected final method.

V. EXPERIMENTAL PROTOCOL

Experiments use Webots R2025a and an e-puck. All steps log sender signals, serialized packets, decoded receiver image, planner state, wheel control, bilateral contact, and physical clearance. Online modules do not receive true obstacle geometry. Physical labels use the frozen 0.037-m footprint and a near-danger boundary of 0.013 m.

Future danger. M9-B is an independent sealed replication with 240 distinct physical configurations: 70 collision, 70 buffered-near, and 100 safe. Its primary endpoint is family-macro 2-s danger AUPRC for $R_1 - R_0$, with a 10,000-replicate paired, episode-stratified bootstrap. $R_2 - R_1$ is descriptive only.

Precursor. Q5 attempts eight new physical families without using precursor outputs in scenario construction. Four produce accepted Safety-Value support: staggered slalom, three-stage weave, opposed gate contraction, and reverse diagonal chicane. The frozen rule is applied once to 19 accepted onsets and 53 event-free episodes. Coverage requires activity strictly before onset; lead is measured from the first active sample in the bounded look-back window.

Closed loop. Q6 comprises 140 new Webots executions over three bounded scheduler generations. The final generation is compared on 20 paired cells (five per positive family) against sealed U_0 and A_0 controls. Primary diagnostics are danger steps and minimum physical clearance. Packet reconciliation and receiver mirroring must pass for each run. No method is selected from Q6; no confirmatory scheduler study or hardware experiment is claimed.

Packet opportunity value. Q6.5 adds five physical families and two development generations. Each paired branch has an identical prefix, three real 24,000-byte packets, and a protected final reserve. Safety utility is lexicographic: collision, then a three-step difference at the frozen 0.12-m danger boundary, then a 1-mm minimum-clearance difference. Generation 2 uses 60 fixed-schedule runs to compare adjacent opportunities at steps 80, 109, 144, and 217. Models use leave-one-family-out folds; scenario identity and evaluator outcomes are forbidden inputs.

VI. RESULTS

A. Future Motion Predicts Physical Danger

M9-B retains all 240 configurations with zero exclusions and passes every support gate. At 2 s, family-macro AUPRC is 0.906311 for R_0 , 0.993730 for R_1 , and 0.999778 for secondary R_2 . The preregistered primary difference $R_1 - R_0$ is 0.087419 with 95% CI [0.077181, 0.098534]. Continuous-clearance MAE falls from 0.021085 m (R_0) to 0.003618 m (R_1) and 0.001829 m (R_2). Thus future rollout is a reliable danger signal, while the additional command schedule explains a smaller secondary gain (Fig. 2).

Earlier closed-loop diagnostics show why this does not directly solve timing. Across six supported traces, the R_1 trigger leads the 25%-cumulative visual, perception, and control-change times by median 2.976, 2.912, and 3.760 s, respectively, and precedes peak control sensitivity by 4.544 s. Risk lead is therefore not communication-value lead.

B. Decision Relevance Can Be Anticipated

Q5 yields 19 accepted onsets in four new positive families. The unchanged precursor covers 16/19 (84.21%); per-family coverage is 75%, 100%, 75%, and 83.3%. Covered-event lead has mean 0.648 s, median 0.576 s, interquartile range 0.432–0.776 s, and range 0.352–1.344 s. The indicator is active in 320/21,840 samples (1.465%). It activates in 11/53 event-free episodes (20.75%). Median available opportunity is 0.224 s, and all 16 covered onsets have at least two consecutive active samples.

The online composite takes 0.126 ms on average, 0.220 ms at the 95th percentile, and 0.578 ms maximum, with 0/1000 misses of the 32-ms control deadline. These results qualify P_t as an efficient cross-scenario precursor to decision relevance, not as evidence of packet utility.

C. Relevance Does Not Guarantee Send Utility

All 140 Q6 runs transmit exactly 72,000 wire bytes; reconciliation and receiver mirroring pass. Relative to A_0 , the final rule reduces mean danger burden by 40.9 steps and raises minimum clearance by 19.75 mm. Nine cells improve, 11 tie, and none are adverse. Reporting only this comparison would be misleading.

Against the stronger U_0 , danger burden still falls by 10.3 steps on average, but minimum clearance falls by 1.24 mm. Outcomes split evenly: 8 improve, 4 tie, and 8 are adverse. Family effects are heterogeneous (Fig. 3). The staggered-slalom family is materially adverse, adding 26.2 danger steps

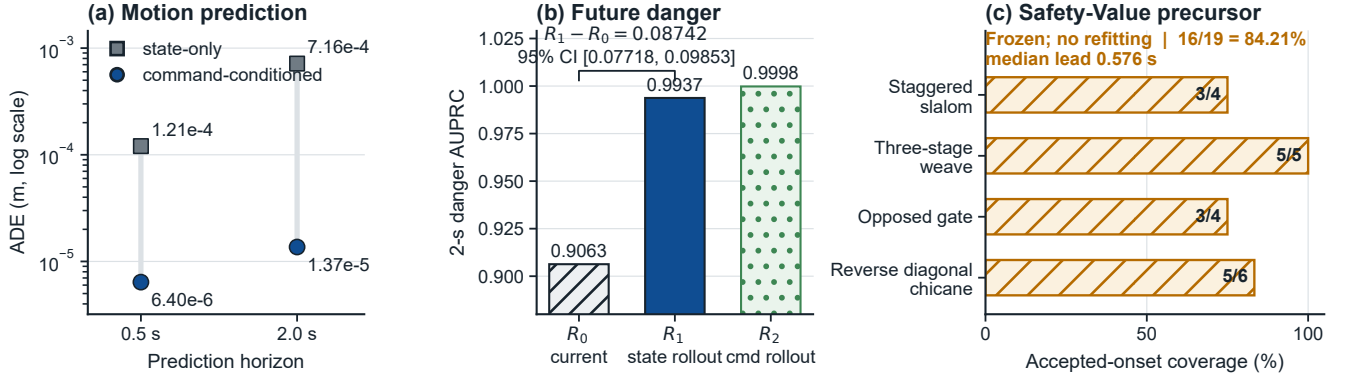


Fig. 2. Evidence chain. (a) Command conditioning improves stable-window ego-motion prediction. (b) On 240 sealed M9-B configurations, future-motion risk R_1 improves 2-s danger discrimination over current-state R_0 ; R_2 is secondary. (c) Without refitting, the frozen precursor covers 16/19 Safety-Value onsets across four new positive families.

TABLE I
CAUSAL EVIDENCE AND CLAIM BOUNDARY.

Stage	Main result	Supported claim
M9-B	$R_1 - R_0$ AUPRC = 0.0874, CI [.0772, .0985]	Future motion predicts danger
Q5	16/19 onsets, 0.576-s median lead, 1.465% active	Decision relevance is anticipatable
Q6 vs A_0	-40.9 danger steps, +19.75 mm; 9/11/0 improve/tie/adverse	Better than reactive baseline
Q6 vs U_0	-10.3 danger steps, -1.24 mm; 8/4/8	No consistent strongest-baseline safety gain
Q6.5	11/15/19 helpful/harmful/neutral; AUPRC 0.628	Utility measurable; no predictor selected

and losing 14.0 mm clearance, whereas three-stage weave improves both measures. No collision-count difference is observed.

A post-hoc grouped diagnostic tested whether a compact learned precursor could repair this heterogeneity. Held-family macro AUPRC was only 0.267 for logistic regression and 0.445 for a tree, with false-warning episode fractions 56.6% and 83.0%, versus 20.8% for the frozen rule. Neither model was integrated.

D. Packet Utility Is Measurable but Not Safely Predicted

Q6.5 intervenes on packet timing rather than treating relevance as a proxy. Thirty matched SEND/HOLD pairs yield 13 helpful, 7 harmful, and 10 neutral opportunities. A runtime-aligned logistic model reaches 0.911 leave-family-out AUPRC, yet its scheduler is worse than U_0 : mean danger increases by 1.73 steps and minimum clearance falls by 1.42 mm. It had learned SEND versus a step-217 fallback, not SEND versus U_0 's step-109 opportunity.

Generation 2 freezes adjacent comparisons before a four-schedule bank. Its 45 opportunities contain 11 helpful, 15 harmful, and 19 neutral labels. Utility is non-monotonic: in each split-gate cell, steps 80 and 109 collide, step 144 avoids collision, and step 217 collides again. The best binary logistic model obtains 0.642 AUPRC with 40% harmful false sends. A preregistered three-class rule reduces harmful false sends

to 20%, but helpful AUPRC is only 0.628 and recall 45.5%. It fails the frozen gate; no method or Q7 study is opened.

VII. DISCUSSION

The experiments support a strict hierarchy: future risk $\neq$ decision value $\neq$ immediate transmission utility. Future risk is predictable seconds before received-image changes strongly affect control. Feasible-set deterioration narrows this gap and anticipates when fresh visual information changes the safety decision. Yet that relevance is only a necessary condition for useful communication. Under a finite packet budget, a relevant frame competes with unknown future observations.

This opportunity cost explains the Q6 pattern. A_0 reacts late enough that a decision-aware precursor can improve it. U_0 , however, preserves a later observation at a fixed time; in sequential obstacles that later packet can be more valuable than the first persistent precursor. The staggered-slalom result is a concrete counterexample to treating P_t as a direct SEND command. The scientific contribution is therefore not a universally safer scheduler, but an experimentally grounded decomposition of what such a scheduler must estimate.

Q6.5 makes the remaining target operational: communication action value is the physical safety difference between adjacent, equal-cost SEND/HOLD branches. It can be measured reproducibly, but the current causal state does not predict it safely across held-out families. Risk magnitude or current decision change alone cannot substitute for unobserved future opportunity structure.

Limitations are substantial. Evidence comes from Webots, one robot model, a controlled visual detector, and a fixed 15-action planner. The exact thresholds in Eq. (8) are system-specific. Q5 is development qualification, while Q6 and Q6.5 are bounded development interventions rather than sealed confirmatory scheduler evaluations. Q6.5 reuses its 15 cells for model development and cross-family replay; it is not independent validation. No method freeze, Q7, formal scheduler result, or real-robot validation exists. These boundaries preclude a claim of closed-loop safety improvement.

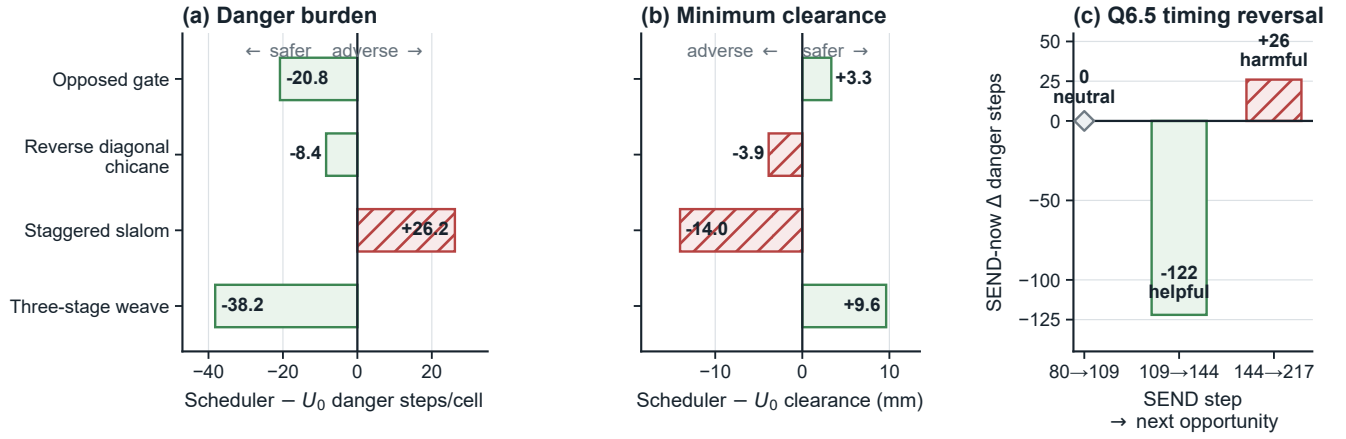


Fig. 3. Matched-cost closed-loop heterogeneity. (a,b) The final development scheduler minus stronger baseline U_0 ; lower danger and higher clearance are safer, while staggered slalom is adverse on both measures. (c) In one frozen Q6.5 cell, adjacent SEND opportunities change from neutral to helpful to harmful, showing that decision relevance does not imply stable packet utility. Every episode transmits exactly 72,000 serialized wire bytes.

VIII. CONCLUSION

We studied when scarce visual updates should be communicated for robot navigation. A sealed 240-configuration evaluation confirms that future-motion risk predicts danger, and a frozen feasibility precursor anticipates decision-relevant visual changes across four new families. Exact matched-wire closed-loop tests then reveal the crucial boundary: relevance improves over a reactive baseline but does not consistently beat uniform timing. Direct matched-byte interventions further show helpful, harmful, neutral, and time-reversing packet opportunities, while bounded causal predictors fail cross-family gates. Risk-aware visual communication must therefore represent future opportunity structure, not only danger and current decision relevance.

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